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Self-Distilled Hierarchical Network for Unsupervised Deformable Image Registration

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***Abstract*— Unsupervised deformable image registration benefits from progressive network structures such as Pyramid and Cascade. However, existing progressive networks only consider the single-scale deformation field in each level or stage and ignore the long-term connection across non-adjacent levels or stages. In this paper, we present a novel unsupervised learning approach named Self-Distilled Hierarchical Network (SDHNet). By decomposing the registration procedure into several iterations, SDHNet generates hierarchical deformation fields (HDFs) simultaneously in each iteration and connects different iterations utilizing the learned hidden state. Specifically, hierarchical features are extracted to generate HDFs through several parallel gated recurrent units, and HDFs are then fused adaptively conditioned on themselves as well as contextual features from the input image. Furthermore, different from common unsupervised methods that only apply similarity loss and regularization loss, SDHNet introduces a novel self-deformation distillation scheme. This scheme distills the final deformation field as the teacher guidance, which adds constraints for intermediate deformation fields on deformation-value and deformation-gradient spaces respectively. Experiments on five benchmark datasets, including brain MRI and liver CT, demonstrate the superior performance of SDHNet over state-of-the-art methods with a faster inference speed and a smaller GPU memory. Code is available at https://github.com/Blcony/SDHNet.**

***Index Terms*— Medical registration, Unsupervised learning, Neural network, Self-distillation.**

I. Introduction

Deformable image registration is an essential task in computer vision, especially in the field of medical image analysis such as medical image segmentation [1] and multi-modality image fusion. Deformable image registration aims to obtain a non-linear correspondence between a pair of images (i.e., a fixed image and a moving image), so that the moving image can be aligned to the fixed image, allowing clinical comparison of images captured from different patients, devices, or times.

Traditional methods [2]–[6] formulate the registration task as an optimization problem for each image pair and demand long running time and high computational cost. Recently, deep learning methods have been utilized in deformable image registration, with promising performance and much less inference time on GPUs. In practice, it is usually hard to obtain the ground-truth deformation field for the supervised registration methods. Therefore, weakly-supervised methods [7] are proposed to utilize the landmarks or segmentation maps to assist the learning of registration. Although weakly-supervised methods alleviate the difficulty of labeling, it still demands human effort. Recently, unsupervised learning has played a vital role in deformable image registration since no labels and landmarks are needed.

U-shape networks [8]–[12] are commonly used to generate the deformation field directly in an unsupervised way, where the moving image is warped only once in the whole procedure. Since U-shape networks (UNets) cannot handle complicated deformations well due to their direct structure, two progressive structures, Pyramid and Cascade, have also been exploited. Pyramid structures [13]–[17] have multiple levels and generate the deformation fields in a coarse-to-fine manner, where the moving image or features in different levels are warped in a progressive procedure. By breaking down a large deformation into small deformations, Cascade structures [18], [19] have multiple stages composed of repetitive sub-networks (e.g., the U-shape network) and generate the deformation field in each stage together with warping the moving image progressively.

Owing to progressive warping operations on the moving image or features, Pyramid and Cascade achieve better registration performance than direct U-shape structures. However, both of them only consider the single-scale deformation field in each level of Pyramid or each stage of Cascade. For Pyramid, each level generates a deformation field at the corresponding scale in a sequential way. Therefore, errors from the coarse levels are hard to be recovered, and small anatomical structures with large deformations may be missed. For Cascade, each stage generates a fine deformation field that lacks explicit coarse information and cannot adaptively focus on deformation with different scales, which is harmful to the efficiency of the registration procedure. Meanwhile, Pyramid or Cascade [13], [15], [16], [18], [19] ignores the long-term connections across non-adjacent levels or stages. Most progressive structures only maintain the connection between adjacent levels or stages through the warped images or features. Although some Pyramid methods [15] strengthen the connection between adjacent levels by concatenating features or deformation fields, the long-term connections across non-adjacent levels have not been exploited well before.

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Moreover, current unsupervised Pyramid or Cascade methods usually rely on the similarity loss and the regularization loss to optimize the whole registration procedure. The similarity loss constrains the reconstruction error between the fixed image and the warped moving image, and the regularization loss constrains the smoothness of the final deformation filed. In fact, there are sequentially generated deformation fields during the progressive registration procedure, and the final deformation field is more accurate than intermediate deformation fields due to the repetitive refinement. Therefore, in addition to the similarity loss, the final deformation field can be a good guidance for the intermediate deformation fields during the training procedure. But how to use the self-contained information such as generated deformation fields to further boost the registration performance is still an open question.

In this paper, we propose a novel unsupervised learning approach for deformable image registration by designing a Self-Distilled Hierarchical Network (SDHNet). SDHNet decomposes the registration procedure into several iterations and generates the deformation fields in a progressive way. In each iteration, to generate hierarchical deformation fields (HDFs) simultaneously, SDHNet adopts a parallel generation module to process hierarchical features extracted from input images. Once obtained the HDFs, SDHNet adopts a contextual adaptive fusion module to estimate a group of pixel-wise adaptive weights conditioned on the HDFs as well as contextual features extracted from the fixed image. Then HDFs can be fused pixel-wisely based on the contextual adaptive weights, resulting in the intermediate deformation field at this iteration. At the same time, to consider the long-term connections across non-adjacent iterations, SDHNet adopts the gated recurrent units (GRUs) in the parallel generation module. Compared with another popular recurrent structure LSTM [20], GRU is computationally more efficient due to the less complex structures and has the comparable performance. The multi-scale hidden states of parallel GRUs pass through different iterations of SDHNet, which build the long-term connections across non-adjacent iterations utilizing the memory characteristics of GRUs. During the whole progressive procedure, intermediate deformation fields are generated sequentially, and the final deformation field can be obtained through the composition of all intermediate ones.

Furthermore, to use the final deformation field for boosting the performance, we propose a novel self-deformation distillation scheme for unsupervised deformable image registration. Specifically, SDHNet treats the final deformation field as the teacher guidance for the sequential intermediate deformation fields, and imposes two constraints on deformation-value space and deformation-gradient space respectively during the training procedure. On the one hand, the intermediate deformation fields can be improved under the guidance of the final deformation field. On the other hand, the improved intermediate deformation fields can be composited to a better final deformation field. It is worth mentioning that this self-distillation scheme does not demand any human labels or landmarks and does not need any pre-trained powerful teacher models. Through this scheme, SDHNet can sufficiently utilize the self-contained information, i.e., the generated deformation fields, which improves the registration accuracy without increasing model size and inference time.

To evaluate the effectiveness and efficiency of SDHNet, we conduct comprehensive experiments on four benchmark datasets including brain MRI and liver CT. Both quantitative and qualitative results demonstrate the superior performance of SDHNet over state-of-the-art methods with a faster inference speed and a smaller GPU memory.

We summarize our main contribution as follows:

- We design a novel unsupervised learning approach SDHNet for deformable image registration, to address the existing problems, such as considering only a single-scale deformation field, lacking long-term connections, and ignoring guidance from the final deformation field.
- We propose the parallel recurrent generation module with the contextual adaptive fusion module, to consider hierarchical deformation fields explicitly and long-term connections between non-adjacent iterations.
- We propose the self-deformation distillation scheme first, by treating the final deformation field as the teacher guidance and intermediate deformation fields as the student.
- We provide comprehensive experiments on five representative datasets including brain MRI and liver CT, to demonstrate the practical effectiveness and efficiency of our SDHNet.

II. Related Work

A. Deformable Image Registration

Traditional methods [2]–[6] regard the registration procedure as an iterative optimization problem and solve it through designed constraints. Depending on the way of modelling the deformation field, there are several groups including elastic-type models [4], free-form deformation model based on the cubic B-Spline [21], statistical parametric mapping [22] and Demons [23]. Many methods have also been proposed to obtain deformation fields within the space of diffeomorphic maps, to guarantee the invertibility properties. These methods include diffeomorphic B-Splines [24], diffeomorphic Demons [25], large deformation diffeomorphic metric [26] and symmetric image normalization (SyN) [3]. Recently, deep neural networks have been applied to unsupervised deformable image registration [27]. VoxelMorph [8], [11] along with its diffeomorphic version [9], [28] adopt U-Net to generate the deformation field directly, which save considerable inference time compared with traditional methods. Then FAIM [10] proposes an explicit anti-folding regularization for U-Net along with modifying the concatenation to element-wise addition.

Learning-based methods mentioned above are all based on a U-shape structure and generate the deformation field in a direct scheme, which means there is a heavy burden for the network optimization when encountering a large deformation. Therefore, progressive structures such as Pyramid and Cascade are proposed to alleviate the difficulty of deformation generation by progressive warping operations on the moving image or features. In the field of Pyramid structures, Dual-PRNet [13] proposes a dual-stream pyramid structure to generate the deformation field in a coarse-to-fine manner. Then Dual-PRNet++ [16] modifies its original structure by introducing

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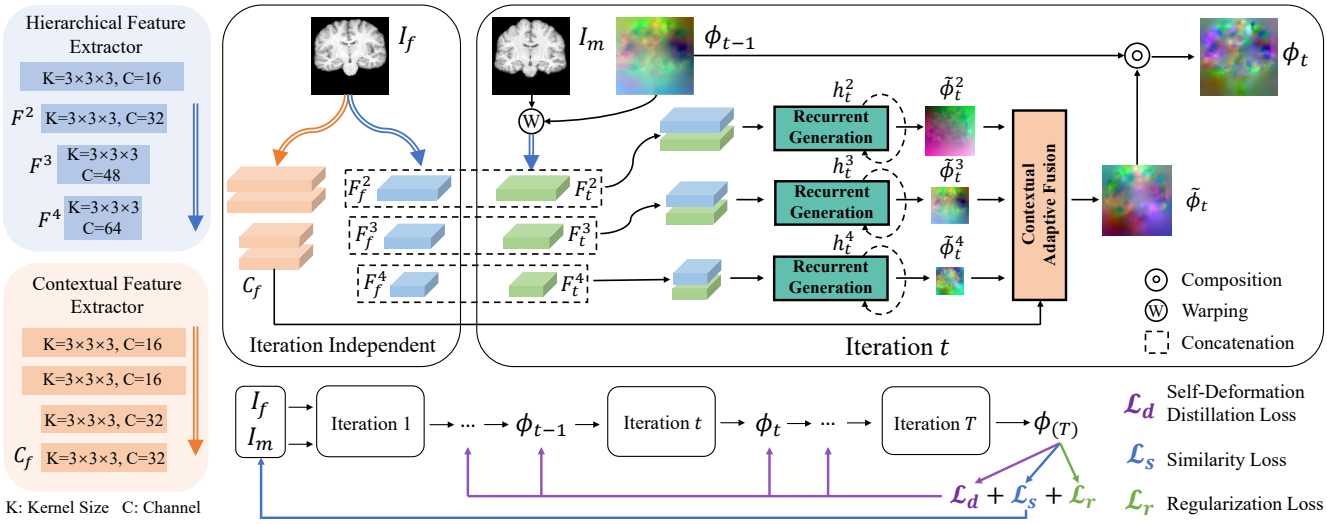


Fig. 1. Overview of our SDHNet. SDHNet decomposes ϕ into T iterations and generates sequential deformation fields $\{\phi_1, ..., \phi_T\}$ where ϕ_T is the final deformation field ϕ. Iteration independent contextual feature C_f and hierarchical fixed features $\{F_f^2, F_f^3, F_f^4\}$ are extracted from the fixed image I_f through the hierarchical feature extractor and contextual feature extractor. At iteration t, SDHNet extracts the hierarchical moving features $\{F_t^2, F_t^3, F_t^4\}$ from the warped moving image $I_m(\phi_{t-1})$. In addition, SDHNet generates hierarchical update deformation fields $\{\tilde{\phi}_t^2, \tilde{\phi}_t^3, \tilde{\phi}_t^4\}$ and hidden states $\{h_t^2, h_t^3, h_t^4\}$ through the parallel recurrent generation module. Then SDHNet fuses $\{\tilde{\phi}_t^2, \tilde{\phi}_t^3, \tilde{\phi}_t^4\}$ as the update deformation field $\tilde{\phi}_t$ through the contextual adaptive fusion module conditioned on C_f. Finally, $\tilde{\phi}_t$ and ϕ_{t-1} can be composed as ϕ_t which is passed into the next iteration with $\{h_t^2, h_t^3, h_t^4\}$. For the self-deformation distillation scheme, the final ϕ can be used as the teacher guidance for the intermediate $\{\phi_1, ..., \phi_{T-1}\}$ by constraining the deformation-value space and deformation-gradient space, respectively.

3D correlation layers and convolutional enhancement while introducing higher GPU memory and longer inference time. LapIRN [15] adopts a image Laplacian pyramid as the input to generate and refine diffeomorphic deformation fields along with the pyramid level. In the field of Cascade structures, DLIR [18] stacks several convolutional neural networks and trains each stage one by one. Then VTN [29] is proposed as a framework that combines rigid registration and deformable registration, and its performance was validated for up to three cascades. Recursive cascaded network [19] takes VTN as its sub-networks, and analyzes the effect of sub-networks in different cascading stages. RTA-Net [30] utilizes a single U-shape network during the training stage and estimates the deformation fields in a cascade-based manner during the inference stage. DiffuseMorph [31] proposes the first diffusion-based registration method conditioned on a pair of moving and fixed images, which cascades the diffusion model with the registration model. Different from the above-existing structures, SDHNet generates HDFs simultaneously while considering long-term connections across non-adjacent iterations.

B. Knowledge Distillation

Knowledge distillation aims to transfer knowledge from a teacher model to a light student model, aiming to improve the performance of the student model without extra storage or computation. Many researchers [32], [33] focus on using it to improve the classification performance of the light student model by distilling knowledge from the output class probabilities of the teacher model. Moreover, there are some knowledge distillation methods have been designed for solving the object detection and semantic segmentation problems [34], [35]. They are proposed to distill interior structural information, which is beneficial for the segmentation or detection problems. Recently, many works have introduced the idea of distillation to medical image analysis. Xing et al. [36] adopted the distillation scheme to solve the overfitting problem in medical image detection. Dian et al. [37] solved medical image segmentation by proposing a comprehensive distillation architecture. In addition, Tran et al. [38] proposed an adversarial learning approach to distill the teacher knowledge into the light student model for deformable image registration.

In addition to improving the light student's performance as described above, knowledge distillation can be used as an efficient scheme to improve the model performance, known as self-knowledge distillation. Zhang et al. [39] introduced the a self-knowledge distillation into the classification problem by distilling the final logits for the intermediate logits. An et al. [40] adopted self-layer context distillation module, which can transfer a correlation map from deep layers to shallow layers. In the field of medical image analysis, Sun et al. [41] and Ni et al. [42] introduce self-knowledge distillation into classification and segmentation respectively, by taking the output of the model itself as the constraint to improve final performance. To the best of our knowledge, there are no methods adopting the self-knowledge distillation for deformable image registration. SDHNet first proposes a novel self-deformation distillation by treating the final deformation field as the teacher guidance and adding constraints for intermediate deformation fields.

III. Method

A. Overview

Fig. 1 illustrates the framework of our proposed SDHNet. Given a pair of the fixed image I_f (we use I to denote the Image) and the moving image I_m, SDHNet aims to generate the (final) deformation field ϕ so that the warped moving image $I_m(\phi)$ can be aligned to I_f. SDHNet decomposes the generation of ϕ into T iterations and generates a sequence of

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deformation fields $\{\phi_t\}_{t=1}^{T-1}$. It is worth mentioning that ϕ_T is the final deformation field ϕ, and there is no extra step between them. We make the final deformation field have two denotations for the convenience of the statements.

Specifically, SDHNet first extracts the contextual feature C_f and hierarchical fixed features $\{F_f^s\}_{s=2}^4$ (we use F to denote the Feature) from the fixed image I_f, and hierarchical moving features $\{F_m^s\}_{s=2}^4$ from the moving image I_m, where s denotes the resolution is $1/2^s$ scale of the full resolution. Then C_f and $\{F_f\}_{s=2}^4$ will participate in the deformation field generation at each iteration. C_f can provide the detailed features of the fixed image which is the target of the registration, therefore giving explicit prior information for the following generation. It is worth mentioning that the extraction of C_f and $\{F_f\}_{s=2}^4$ is iteration independent, which means they are extracted only once in the whole procedure and do not change in different iterations. This design alleviates the repetitive usage of the fixed image, reducing the network's inference burden.

At the beginning of the iteration as the iteration 1, the input is the hierarchical fixed features $\{F_f^s\}_{s=2}^4$ with the moving features $\{F_m^s\}_{s=2}^4$. The input is first fed into the parallel recurrent generation module to simultaneously obtain hierarchical update deformation fields $\{\tilde{\phi}_1^s\}_{s=2}^4$. Then, we feed $\{\tilde{\phi}_1^s\}_{s=2}^4$ and C_f into the contextual adaptive fusion module to generate the update deformation field $\tilde{\phi}_1$ at the iteration 1, which is passed into the iteration 2.

The following procedure until the iteration T is the same, and we take the iteration t as an example. At the iteration t, given the input deformation field ϕ_{t-1} from the last iteration $t-1$, SDHNet extracts the hierarchical warped moving features $\{F_t\}_{s=2}^4$ from the warped moving image $I_m(\phi_{t-1})$. $\{F_f\}_{s=2}^4$ and $\{F_t^s\}_{s=2}^4$ are the inputs for this iteration, and the following process is the same as the iteration 1. The output is the update deformation field $\tilde{\phi}_t$ at this iteration. Unlike directly passing $\tilde{\phi}_t$ into the next iteration like iteration 1, we composite $\tilde{\phi}_t$ with the input deformation field ϕ_{t-1} to obtain the latest deformation field ϕ_t as $\phi_t = \phi_{t-1} \circ \tilde{\phi}_t$ where $\circ$ denotes the composition operation, and ϕ_t is passed to the next iteration $t+1$.

Furthermore, we propose a novel self-deformation distillation scheme to boost the registration performance. This scheme treats the final deformation field ϕ as the teacher guidance and adds two constraints for intermediate deformation fields $\{\phi_t\}_{t=1}^{T-1}$ on the deformation-value space and deformation-gradient space respectively. Additionally, the stationary velocity field (SVF) can be readily integrated into SDHNet, aiming to preserve the diffeomorphic property of deformation fields. In the following sections, we elaborate on the crucial components of SDHNet in detail, including feature extraction, parallel recurrent generation, contextual adaptive fusion, and self-deformation distillation. We clarify the diffeomorphic version of SDHNet at the end.

B. Hierarchical & Contextual Feature Extraction

There are two multi-scale encoder structures in our SDHNet: a hierarchical feature extractor and a contextual feature extractor. We basically follow the design of the multi-scale encoder in the representative work VoxelMorph [8]. The hierarchical feature extractor has four convolutional layers, each of which has a 3×3×3 kernel size with a stride of 2. Hierarchical feature extractor is used for extracting hierarchical (fixed/moving) features The 1/4 scale features F^2, 1/8 scale features F^3 and 1/16 scale features F^4 are the output from the second, third and fourth convolutional layers, respectively. The channel dimensions of F^2, F^3 and F^4 are 32, 48 and 64, respectively. The contextual feature extractor has four convolutional layers with 3×3×3 kernels, while the first and third layers have a stride of 2, and the second and the fourth layer has a stride of 1. The contextual feature C_f is the output from the fourth layer with a 1/4 scale and a 32-channel dimension. We provide the detailed structure of the hierarchical feature extractor and contextual feature extractor in the left part of Fig. 1, including the kernel size, channel dimension and number of convolutional layers. We mark the corresponding features for the extractors.

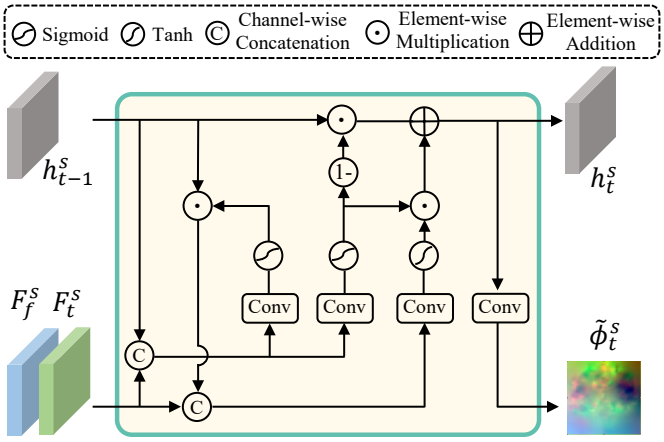


Fig. 2. The structure of the recurrent generation module.

C. Parallel Recurrent Generation

Given the input fixed features $\{F_f^s\}_{s=2}^4$ and the moving features $\{F_t^s\}_{s=2}^4$ at the iteration t, SDHNet divides them into three groups as $\{F_m^s, F_t^s\}_{s=2}^4$ according to the resolution scale. Then three individual generation modules are applied for different groups to generate hierarchical update deformation fields $\{\tilde{\phi}_t^s\}_{s=2}^4$. For each individual generation module, it is only responsible for processing the corresponding feature to obtain a single-scale deformation field, and cross-scale information does not pass between different generation modules. We think this design can help the generation module focus on specific single-scale information. It is worth noting that there is no coarse-to-fine procedure so that hierarchical update deformation fields can be generated simultaneously in a parallel scheme, which is beneficial for the efficiency of registration. Moreover, SDHNet adopts a modified GRU module as the generation module to consider long-term connections across non-adjacent iterations, where fully connected layers are replaced with convolutions. As shown in Fig. 2, it processes the input features F_f^s and F_t^s along with the learned hidden state h_{t-1}^s from last iteration, and outputs the update hidden state h_t as following,

$$z_t^s = \sigma\left(\mathrm{Conv}_{3\times3\times3}\left(\left[h_{t-1}^s, F_f^s, F_t^s\right], W_z^s\right)\right), \quad (1)$$

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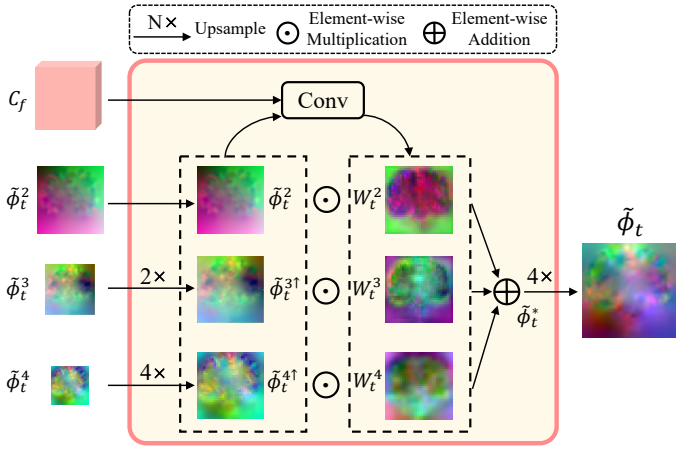


Fig. 3. The structure of the contextual adaptive fusion module.

$$r_t^s = \sigma\left(\mathrm{Conv}_{3\times3\times3}\left(\left|h_{t-1}^s, F_f^s, F_t^s\right|, W_r^s\right)\right), \tag{2}$$

$$\tilde{h}_t^s = \tanh\left(\mathrm{Conv}_{3\times3\times3}\left(\left[r_t^s \odot h_{t-1}^s, F_f^s, F_t^s\right], W_h^s\right)\right), \tag{3}$$

$$h_t^s = (1 - z_t^s) \odot h_{t-1}^s + z_t^s \odot \tilde{h}_t^s, \tag{4}$$

where $\mathrm{Conv}_{3\times3\times3}$ means one 3×3×3 convolutional layer and W denotes corresponding learnable kernel parameters in the convolutional layer. Then the hidden state h_t^s is passed through two convolutional layers to generate the update deformation field $\tilde{\phi}_t^s$. The reason that the long-term connection across non-adjacent iterations can be considered is depicted in Eq. 4, where h_t^s involves all previous hidden states $\{h_1^s, ..., h_{t-1}^s\}$ in an adaptive summation. Therefore, the learned hidden state is able to help the network choose which iteration should be strengthened or weakened, and the generation of $\tilde{\phi}_t^s$ can be affected by all previous ones.

D. Contextual Adaptive Fusion

Given hierarchical update deformation fields $\{\tilde{\phi}_t^s\}_{s=2}^4$ at the iteration t, SDHNet demands to fuse them as the update deformation field $\tilde{\phi}_t$. The scale of $\{\tilde{\phi}_t^2, \tilde{\phi}_t^3, \tilde{\phi}_t^4\}$ corresponds to {1/4, 1/8, 1/16}. Due to the scale difference, we first upsample $\tilde{\phi}_t^3$ to $\tilde{\phi}_t^{3\uparrow}$ with the same scale as $\tilde{\phi}_t^2$ through the trilinear interpolation. Following the same way, we can easily obtain $\tilde{\phi}_t^{4\uparrow}$. To fuse $\{\tilde{\phi}_t^2, \tilde{\phi}_t^{3\uparrow}, \tilde{\phi}_t^{4\uparrow}\}$, a straightforward method is to apply an average summation. But average summation is not learnable and flexible enough. On the one hand, the impact of $\{\tilde{\phi}_t^2, \tilde{\phi}_t^{3\uparrow}, \tilde{\phi}_t^{4\uparrow}\}$ varies in different iterations and constant weights in an average summation cannot distinguish this difference. On the other hand, because diverse anatomical structures have different deformation degrees, different spatial position in the same deformation field demands different treatments, while constant weights cannot satisfy.

As a result, SDHNet adopts the contextual adaptive fusion module to assign pixel-wise adaptive weights $\{W_t^2, W_t^3, W_t^4\}$ to $\{\tilde{\phi}_t^2, \tilde{\phi}_t^{3\uparrow}, \tilde{\phi}_t^{4\uparrow}\}$, respectively. In the process of generating weights, rather than designing complicated operations to consider the difference of hierarchical deformation fields, we just utilize $\{\tilde{\phi}_t^2, \tilde{\phi}_t^{3\uparrow}, \tilde{\phi}_t^{4\uparrow}\}$ themselves as the input for two convolutional layers. Meanwhile, to take advantage of the prior contextual information from the fixed image directly, we use the contextual feature C_f additional as a complementary input, aiming to help the weights focus on the fixed image's explicit content. To keep the fused deformation field in a reasonable deformation range, we constrain the summation of the adaptive weights $\{W_t^2, W_t^3, W_t^4\}$ to be equal to 1 in a pixel-wise manner. Specifically, we denote the output of two convolutional layers as the temporal weights $\{\tilde{W}_t^2, \tilde{W}_t^3, \tilde{W}_t^4\}$. Then the utilized pixel-wise adaptive weights $\{W_t^2, W_t^3, W_t^4\}$ are the softmax normalization results of the temporal weights $\{\tilde{W}_t^2, \tilde{W}_t^3, \tilde{W}_t^4\}$ in a pixel-wise manner as

$$W_t^s(i,j) = \frac{exp(\tilde{W}_t^s(i,j))}{\sum_{s=2}^4 exp(\tilde{W}_t^s(i,j))}, \tag{5}$$

where (i, j) is the spatial pixel position, and $s \in \{2, 3, 4\}$ is the weight scale. By the normalization of the softmax function, we can guarantee the constraint about the summation of the adaptive weights $\{W_t^2, W_t^3, W_t^4\}$ as 1 in a pixel-wise manner. Then, as shown in Fig. 6, the fusion process can be depicted as following,

$$\tilde{\phi}_t^* = W_t^2 \odot \tilde{\phi}_t^2 + W_t^3 \odot \tilde{\phi}_t^{3\uparrow} + W_t^4 \odot \tilde{\phi}_t^{4\uparrow}, \tag{6}$$

where $\odot$ is the element-wise multiplication. We upsample $\tilde{\phi}_t^*$ to the full resolution as $\tilde{\phi}_t$ by the trilinear interpolation.

E. Self-Deformation Distillation

Due to the repetitive refinement, the final deformation field ϕ can obtain better registration accuracy than intermediate ones $\{\phi_t\}_{t=1}^{T-1}$, so ϕ can be a good guidance for $\{\phi_t\}_{t=1}^{T-1}$. To use the final deformation field for further boosting the performance, we propose a self-deformation distillation scheme. This scheme treats the final deformation field ϕ as the teacher guidance, and adds a distillation loss $\mathcal{L}_d$ for the intermediate deformation fields $\{\phi_t\}_{t=1}^{T-1}$. Concretely, $\mathcal{L}_d$ consists of two terms, one is $\mathcal{L}_{d-v}$ based on the deformation-value space, and another is $\mathcal{L}_{d-g}$ based on the deformation-gradient space. The loss $\mathcal{L}_{d-v}$ just constrains the mean square error between ϕ and $\{\phi_t\}_{t=1}^{T-1}$ as following,

$$\mathcal{L}_{d-v} = \sum_{t=1}^{T-1} \lambda^{T-1-t}(\phi - \phi_t)^2, \tag{7}$$

where λ^{T-1-t} is the balance weight, meaning the distillation degree for the intermediate deformation fields is different. In the forward procedure, when the deformation field is generated later, it is more accurate and closer to the final deformation field, and vice versa. Therefore, when we assign the distillation weight, we maintain the relationship among the intermediate deformation fields. We assign relatively small weights for early intermediate deformation fields and relatively large weights for later ones. Inspired from the supervised optical flow network RAFT [43], we choose the exponential function and set the power of the exponential λ function in the range of [0, 1] to guarantee that the function is monotonically increasing.

Through L_{d-v}, we enable the intermediate deformation fields to learn more directionally guided by the final deformation field. In other words, under the guidance from the final deformation field, intermediate deformation fields can consistently contribute to the accurate registration performance and

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these in the wrong deformation direction can be constrained. At the same time, the improved intermediate fields can be composited as the better final deformation field at the next training step, which constructs a positive circuit. Furthermore, besides the deformation-value difference, $\mathcal{L}_{d-g}$ is used to constrain the difference between ϕ and $\{\phi_t\}_{t=1}^{T-1}$ on the spatial gradient space. Spatial gradient is usually used to measure the smoothness of the deformation field [8], and we find taking advantage of the smoothness prior can be beneficial for the self-distillation. We think the reason is that, the minimizing of $\mathcal{L}_{d-v}$ itself is not so easy for the network optimization, and the smoothness prior of deformation fields can regarded a helpful complementary to $\mathcal{L}_{d-v}$ optimization by introducing $\mathcal{L}_{d-g}$. The specific formulation of $\mathcal{L}_{d-g}$ is shown as

$$\mathcal{L}_{d-g} = \sum_{t=1}^{T-1} \lambda^{T-1-t} (\nabla(\phi) - \nabla(\phi_t))^2, \tag{8}$$

where ∇ means the spatial gradient operation. The whole distillation loss L_d is depicted as

$$\mathcal{L}_d = \mathcal{L}_{d-v} + \mathcal{L}_{d-g}. \tag{9}$$

When optimizing the model parameters, the back-propagation gradient of $\mathcal{L}_d$ will not pass through ϕ.

Additionally, SDHNet still has two other regular unsupervised loss constraints. The first one is the similarity loss $\mathcal{L}_s$ between the fixed image I_f and the warped moving image $I_m(\phi)$, and the second one is the regularization loss $\mathcal{L}_r$ on the update deformation field $\tilde{\phi}_t$ in each iteration. Specifically, we adopt the negative local cross-correlation loss as the similarity loss and the total variation loss (the differences of spatial gradients of the voxels in the deformation field) as the regularization loss following [29]. Self-deformation distillation loss is simultaneously evoked with the similarity loss and regularization loss after computing the final deformation field, and the whole loss constraint L is shown as as

$$\mathcal{L} = \mathcal{L}_s + \alpha \mathcal{L}_r + \beta \mathcal{L}_d, \tag{10}$$

where α is the regularization weight and β is the distillation weight. All three losses L_d, L_s and L_r are compu It is worth noting that SDHNet is trained in an unsupervised end-to-end way without the usage of any human labels and landmarks.

F. Differomorphic Version of SDHNet

In addition to generating deformation fields directly, we further provide a diffeomorphic version of SDHNet which can preserve the diffeomorphic property of deformation fields explicitly. Specifically, due to the flexibility of SDHNet, SVF can be adopted easily to model deformation fields in SDHNet. SVF defines the deformation field ϕ as $\frac{\partial \phi^{(k)}}{\partial t} = v(\phi^{(k)})$, subject to $\phi^{(0)} = Id$. Therefore, SDHNet should adopt the parallel generation module to output the hierarchical velocity fields $\{v_t^s\}_{s=2}^4$. Then, the contextual adaptive fusion module takes C_f and $\{v_t^s\}_{s=2}^4$ as the input and obtain the fused velocity filed v_t^* following the same procedure. Finally, we exploit the scaling and squaring [44] operation to integrate v_t^* over unit time with setting time step K as 7 to obtain $\tilde{\phi}_t^*$, and upsample it as the update deformation field $\tilde{\phi}_t$. After that, the subsequent procedure can be processed in the same way.

IV. Experiments

A. Datasets

We conduct experiments on five representative datasets for a comprehensive comparison, including three brain MRI datasets LPBA, OASIS and Mindboggle-101, and two liver CT datasets SLIVER and LSPIG.

The LPBA [45] dataset contains 40 scans with corresponding segmentation ground truth of 56 anatomical structures. Following [19], we just utilize LPBA for validation and testing and choose ADNI [46], ABIDE [47], ADHD [48] dataset for training. We adopt the atlas-based strategy which means there is only one fixed image (termed atlas) and every moving image needs to be aligned with it. In detail, there are 66, 1287, 949 scans in ADNI, ABIDE and ADHD respectively for training, and we split LPBA into 9 scans for validation, 30 scans for testing and one scan as the atlas. We carry out standard preprocessing steps for the data following [19], [29]. Every scan is resampled to $128 \times 128 \times 128$ voxels after cropping around the brain. Besides, we provide another version of the LPBA dataset (LPBA-v2) to conduct comprehensive experiments. The resolution of this LPBA-v2 dataset is $160 \times 192 \times 160$ after preprocessing steps such as center-cropped following [16]. We split the LPBA-v2 dataset into 25, 5, 10 images for training, validation and test sets, and conduct the subject-to-subject registration experiments.

The OASIS [49] dataset is broadly used to evaluate registration methods. We adopt the Neurite OASIS [50] version, whose preprocessing steps are the same as HyperMorph [51]. It includes 414 scans with a segmentation annotation on 35 issues, and we split it into 380, 14 and 20 scans as the training, validation and testing dataset respectively. We adopt the subject-to-subject registration strategy, meaning the fixed image is arbitrary and the model is tested on 20×19 pairs. The resolution for scans of this dataset is $160\times$ 192 $\times 224$.

The Mindboggle-101 [52] is a commonly used dataset which focuses on the cerebral cortex regions. We follow [53] to take the NKI-RS-22 subset and the NKI-TRT-20 subset as our training sets (42×41 pairs), and split MMRR-21 subset as our validation set (8×7 pairs) and test set (12×11 pairs). Meanwhile, We conduct standard preprocessing steps such as center-cropping and min-max normalization following [53]], and the resolution of the processed scans is $192 \times 192 \times 192$.

The SLIVER [54] dataset contains 20 scans with liver segmentation ground truth. Following [19], we use SLIVER for validation and testing while choosing MSD [55] and BFH [29] dataset for training. In detail, we select 933 scans of which most likely include livers in MSD, and 92 scans of BFH as our training data. We adopt the subject-to-subject registration strategy. So we split SLIVER into 5 scans (5×4 pairs) as the validation data and 15 scans (15×14 pairs) as the testing data. The resolution for scans of this dataset is $128 \times 128 \times 128$.

The LSPIG [19] dataset is provided by the First Affiliated Hospital of Harbin Medical University, containing 17 pairs of CT scans from pigs, along with liver segmentation ground truth. Each pair comes from one pig with and without 13 mm Hg pneumoperitoneum pressure. The resolution for scans of this dataset is $128 \times 128 \times 128$. Considering the species

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difference, LSPIG is suitable for evaluating the generalization ability of different models. Therefore, we just utilize the LSPIG as the testing data and apply the model trained on the MSD and BFH to it directly.

B. Metrics

We evaluate the registration performance in two aspects, similarity property and diffeomorphic property. For similarity property, we adopt the Dice score as our quantitative metric, which is designed to measure the overlap degree of two regions. The higher Dice score depicts a better similarity property. When evaluating scans containing multiple anatomical structures, we compute Dice scores separately and take an average of them to get the final Dice score following [8]. We adopt the 95% maximum Hausdorff distance (HD95) as an additional similarity metric. HD95 is a commonly used to measure the degree of the estimation errors between two corresponding anatomical ares ourlines [56]. The lower HD95 value is, the closer the corresponding locations between the fixed and warped image are. For diffeomorphic property, we adopt Folds as our quantitative metric following Voxelmorph-*diff* [28]. Specifically, given a deformation field ϕ, we calculate the Jacobian determinant of ϕ ($|J(\phi)|$), and we define the percentage of the negative value of entire volume in Jacobian determinant ($|J\phi| < 0$) as the Folds. Lower Folds depict the better diffeomorphic property. A region of negative determinants indicates that the one-to-one mapping has been lost [2]. Moreover, we test running time and GPU memory on OASIS where the resolution is $160\times$ 192 $\times 224$ during the inference time. We use an Intel Xeon E5-2650 V4 CPU for the traditional methods and an NVIDIA TITAN XP GPU for the learning-based methods.

C. Baseline Methods

We implement three traditional methods and seven learning-based methods as our comparison methods. For traditional methods, following the implementation and commands in cascade VTN [19], [29], we implement SyN [3] by the popular ANTs software package [57] and B-Spline [21] by Elastix [6]. Following the classical baseline [58] and LapIRN [15], we implement the traditional method Demons [25], [59], [60]. Both of them are top-performing traditional algorithms and are widely used in research and clinic. For learning-based methods, we first implement VoxelMorph [8] along with its diffeomorphic version VoxelMorph-*diff* [9] and a single stage VTN [29], which are based on U-shape structures. For Pyramid structures, we implement the representative methods including Dual-PRNet [13], LapIRN [15] along with its displacement version LapIRN-*disp*. We implement the recent state-of-the-art method Dual-PRNet++ [16] for a comparison. For Cascade structures, we choose the VTN cascade structures [19] which have shown the state-of-the-art registration performance.

D. Implementation

SDHNet is implemented on PyTorch and we set the batch size as 4 on 4 cards of NVIDIA TITAN XP GPU during the training time. We utilize the Adam optimizer with parameters β_1 as 0.9, β_2 as 0.99. We run SDHNet for 10^5 steps in the training stages and set the learning rate as 10^{-4} initially, which halves after 6×10^4 and 8×10^4 iteration. We set the number of the iteration T as 6 in SDHNet in default. Based on the results on the validation set, the parameters λ, α and β can be set as 0.8, 0.5 and 0.1 for all datasets. Following [29], we embed the same affine network on top of each network for brain and liver datasets except the pre-affined OASIS, LPBA-v2 and Mindboggle-101 dataset, so that it can predict an affine transformation to align the input pair rigidly before the deformable transformation. Meanwhile, we set the modules in different iterations to have the individual parameters, and the weight-sharing scheme is ablated in Section V-B.4. We also implement the diffeomorphic version of SDHNet as SDHNet-*diff* according to Section III-F, and the default setting is the same as SDHNet except α is set as 0.7.

V. Results

A. Comparison with Baseline Methods

We compare SDHNet with traditional methods, U-shape methods, Pyramid methods, and Cascade methods on LPBA, LPBA-v2, OASIS, Mindboggle-101, and SLIVER datasets, in terms of Dice, HD95 and Folds along with the inference time and GPU memory in Table I, Table II and Table III. We provide the registration performance on each anatomical structures in the OASIS dataset in Fig. 5. Then we evaluate the generalization performance on the LSPIG dataset in Table III. Finally, we provide the statistical analysis on all datasets.

1) Comparison with Traditional Methods: Compared with traditional methods SyN, B-Spline and Demons, SDHNet achieves the best registration accuracy with a fast inference speed and a small GPU memory on the LPBA, LPBA-v2, OASIS, Mindboggle-101, and SLIVER datasets. Specifically, SDHNet outperforms Demons on both LPBA and LPBA-v2 datasets in Table I, along with the Liver CT dataset such as SLIVER in Table III. In addition, our SDHNet improves the Dice score about 6% on the OASIS dataset, 8% on the Mindboggle-101 dataset in Table II. In terms of distance metric, SDHNet achieves the best HD95 performance, by reducing the metric from 1.794 to 1.423 on the OASIS dataset and from 5.081 to 4.468 on the Mindboggle-101 dataset. SDHNet reduces the inference time from hundreds of seconds to less than a second (2683s/378s/214s→0.32s) substantially.

2) Comparison with U-shape Methods: Compared with U-shape structures such as VTN, our SDHNet achieves a consistent improvement in terms of Dice score compared with VTN, such as 3.3% and 4.5% improvement on the LPBA and LPBA-v2 datsets, as shown in Table I. SDHNet improves the Dice score from 76.0% to 81.5% on the OSAIS dataset and from 50.9% to 63.5% on the Mindboggle-101 dataset in Table II, while performing the best HD95 performance with about 0.5 and 0.9 reduction respectively. For the liver CT such as SLIVER dataset, SDHNet still outperforms VTN significantly in Table III. It is worth mentioning that, though SDHNet improves the Dice score compared with VTN, it only demands a similar inference time and GPU memory, which demonstrates the efficient design of SDHNet. In addition to VTN, VoxelMorph is also limited by the direct U-shape

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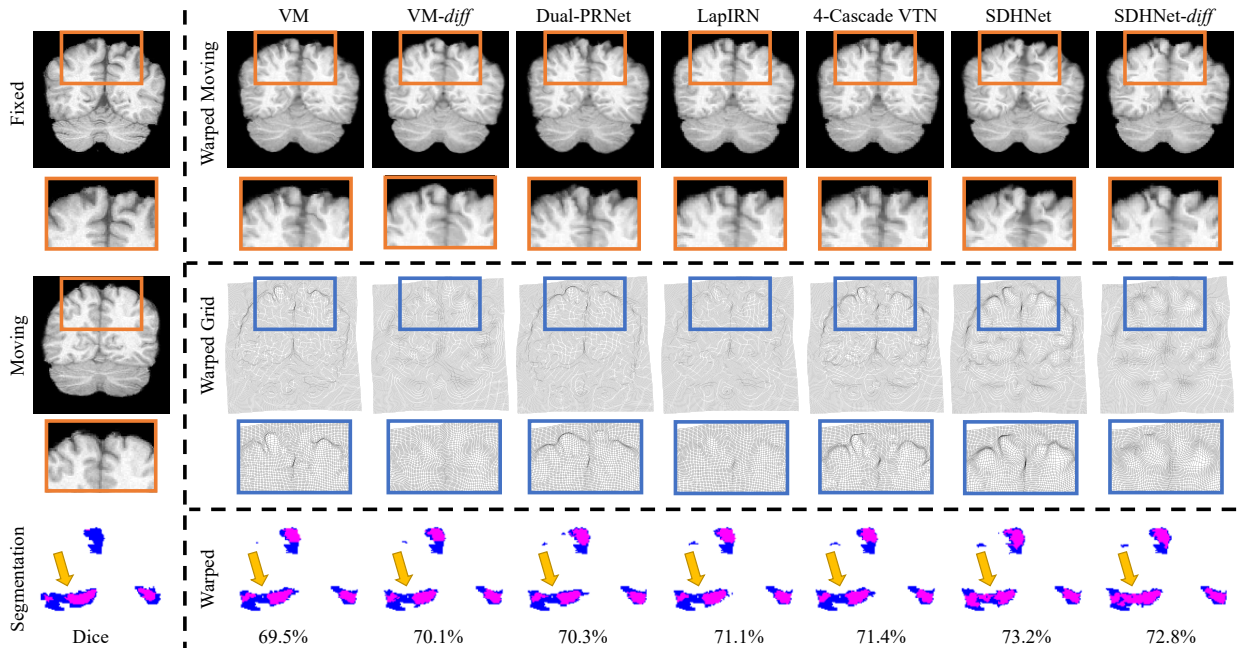


Fig. 4. Visualization of an example in the brain dataset LPBA. Left: the fixed image, the moving image and three segments of representative anatomical structures. Right: the first row shows the corresponding warped moving images, the second row shows the warped grids to visualize diffeomorphic property, and the third row shows the anatomical structures of the fixed image and the warped moving images. Magenta indicates the overlaps of them while blue indicates the rest parts. Yellow arrows indicate a location to distinguish performance. Dice score of the 3D LPBA exmaple is shown below segmentation.

TABLE I

COMPARISON RESULTS ON THE LPBA DATASET AND LPBA-V2 DATASET. FOLDS ARE PRESENTED IN E-NOTATION (E.G., 1e-2=0.01). THE STANDARD DEVIATION IS IN PARENTHESES. **BOLD** FONT MEANS THE BEST AND UNDERLINE MEANS THE SECOND-BEST.

Methods	128 × 128 × 128		160 × 192 × 160	
	Dice (%)	Folds (%)	Dice (%)	Folds (%)
ANTs SyN	70.4 (1.5)	**0**	72.9 (2.1)	**0**
Elastix B-Splines	67.9 (1.3)	**0**	71.1 (1.7)	**0**
Demons	70.7 (1.4)	**0**	73.4 (1.6)	**0**
VoxelMorph	69.0 (1.5)	9.1e-3	71.5 (1.9)	1.9e-2
VoxelMorph-*diff*	69.7 (1.5)	**0**	72.3 (1.6)	**0**
VTN	69.3 (1.4)	5.9e-3	72.1 (1.7)	1.6e-2
Dual-PRNet	69.7 (1.3)	4.5e-3	73.2 (2.0)	1.2e-2
LapIRN-*disp*	70.9 (1.5)	7.7e-3	75.2 (1.6)	1.7e-2
LapIRN	70.6 (1.5)	**0**	73.6 (1.6)	**0**
Dual-PRNet++	71.2 (1.3)	3.0e-3	75.5 (1.9)	9.6e-3
2-cascade VTN	69.8 (1.4)	4.4e-3	74.1 (1.6)	1.2e-2
3-cascade VTN	70.4 (1.3)	3.5e-3	74.9 (1.5)	6.0e-3
4-cascade VTN	70.9 (1.5)	2.7e-3	75.8 (1.5)	3.4e-3
SDHNet	**72.6** (1.2)	1.5e-3	**76.9** (1.6)	1.8e-3
SDHNet-*diff*	72.2 (1.3)	**0**	76.4 (1.5)	**0**

structure. Compared with VoxelMorph, SDHNet improves the Dice score about 5% on OASIS, 8% on Mindboggle-101 and 4% on SLIVER. For the diffeomorphic version, SDHNet-*diff* has lower Folds while obtaining a higher Dice score compared with VoxelMorph-*diff* on all four datasets.

3) Comparison with Pyramid Methods: As shown in Table I, Table II and Table III, compared with Pyramid methods, SDHNet achieves the best Dice score and best Folds on all four datasets. On the LPBA and LPBA-v2 datasets, SDHNet outperforms other methods clearly, as improving the Dice score from 69.7% to 72.6% and from 73.2% to 76.9% compared with Dual-PRNet. On the OASIS dataset, SDHNet improves the Dice score from 73.4% to 81.5% and reduces the HD95 from 1.831 to 1.423 compared with Dual-PRNet, while improving the Dice score from 80.2% to 81.5% and reducing the HD95 from 1.695 to 1.423 compared with LapIRN-*disp*. Even compared with the recent state-of-the-art Dual-PRNet++, SDHNet still achieves an obviously better performance as improving the Dice score from 80.4% to 81.5%. On the Mindboggle-101 dataset, SDHNet also achieves the best registration performance with at least 1.5% improvement for Dice score. At the same time, SDHNet also achieves the best registration performance on the SLIVER dataset. It is worth mentioning that SDHNet demands much less inference time (0.91s→0.32s) and GPU memory (4542MB→2959MB) compared with Dual-PRNet++. Additionally, SDHNet-*diff* also achieves the second-best Dice score compared with all Pyramid methods on all four datasets, and it has a better diffeomorphic performance than SDHNet.

4) Comparison with Cascade Methods: We cascade up to 4 stages for VTN following [19], and show the comparison results in Table I, Table II and Table III. Although cascade methods achieves a better performance than the direct U-shape methods, they demand more inference time and GPU memory which is not friendly in clinic scenarios. However, SDHNet obtains the best performance even compared with heavy cascade methods with a much lighter model. SDHNet achieves the best registration accuracy on the LPBA and LPBA-v2 datasets with a 72.6% and 76.9% Dice score. At the same time, SDHNet achieves the best registration accuracy on the OASIS dataset with a 81.5% Dice score (1.423 HD95), which is better than 2-cascade VTN with a 78.4% Dice score (1.556 HD95) and 3-cascade VTN with a 80.2% Dice

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TABLE II

COMPARISON RESULTS WITH THE TRADITIONAL METHODS, U-SHAPE STRUCTURES, PYRAMID STRUCTURES, CASCADE STRUCTURES, AND SDHNET. FOLDS ARE PRESENTED IN E-NOTATION (E.G., $1e$-2=0.01). TIME AND GPU ARE TESTED ON THE OASIS DATASET DURING THE INFERENCE STAGE.THE STANDARD DEVIATION IS IN PARENTHESES. * MEANS INFERENCE TIME IS TESTED ON CPU WHILE OTHERS ARE TESTED ON GPU. **BOLD** FONT MEANS THE BEST AND UNDERLINE MEANS THE SECOND-BEST.

Methods	OASIS			Mindboggle-101			Time	GPU
	Dice (%)	HD95	Folds (%)	Dice (%)	HD95	Folds (%)	(s)	(MB)
ANTs SyN	74.8 (2.7)	2.106 (0.299)	**0**	54.9 (2.1)	5.126 (0.338)	**0**	2683	-
Elastix B-Splines	74.6 (3.2)	2.039 (0.301)	3.9e-2	54.4 (2.1)	5.878 (0.507)	2.2e-2	378*	-
Demons	75.2 (2.4)	1.794 (0.284)	**0**	55.2 (1.7)	5.081 (0.419)	**0**	214*	-
VoxelMorph	75.6 (2.8)	1.732 (0.296)	4.6e-2	55.6 (1.8)	5.023 (0.402)	4.0e-2	0.19	4120
VoxelMorph-*diff*	75.9 (2.4)	1.768 (0.276)	**0**	55.2 (1.5)	5.011 (0.349)	**0**	0.41	4289
VTN	72.5 (3.2)	1.913 (0.274)	4.5e-2	50.9 (1.4)	5.158 (0.388)	7.5e-3	0.34	2832
Dual-PRNet	73.4 (2.4)	1.831 (0.244)	5.7e-2	52.1 (1.1)	4.818 (0.326)	2.2e-2	0.24	2542
LapIRN-*disp*	80.2 (2.3)	1.695 (0.287)	3.4e-2	61.6 (1.3)	4.927 (0.380)	2.8e-2	0.43	4906
LapIRN	79.5 (2.1)	1.709 (0.258)	**0**	60.6 (1.3)	5.102 (0.358)	**0**	0.86	5182
Dual-PRNet++	80.4 (2.2)	1.628 (0.270)	2.9e-2	61.8 (1.2)	4.704 (0.351)	1.5e-2	0.91	4542
2-cascade VTN	78.5 (2.4)	1.556 (0.228)	9.0e-3	57.1 (1.1)	4.841 (0.366)	7.2e-3	0.62	3320
3-cascade VTN	80.2 (2.4)	1.675 (0.243)	3.6e-3	60.0 (1.1)	4.639 (0.360)	5.8e-3	0.94	3642
4-cascade VTN	80.5 (2.0)	1.597 (0.240)	1.8e-3	61.2 (1.2)	4.526 (0.362)	3.7e-3	1.14	3954
SDHNet	**81.5** (2.2)	**1.423** (0.246)	5.4e-4	**63.5** (1.3)	**4.468** (0.365)	4.7e-3	0.32	2959
SDHNet-*diff*	81.2 (2.1)	1.449 (0.265)	**0**	63.0 (1.1)	4.491 (0.323)	**0**	0.47	3264

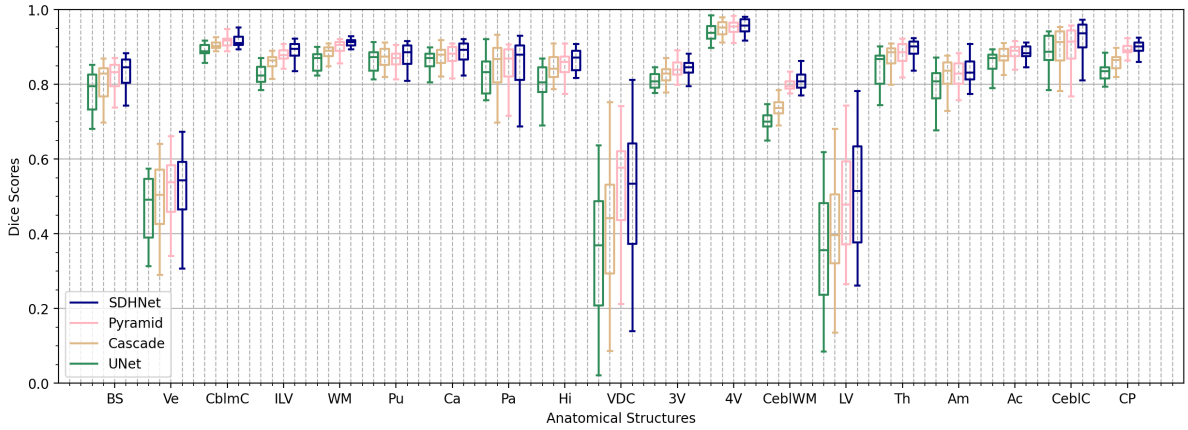


Fig. 5. Boxplots about the Dice scores of all anatomical structures on OASIS for UNet (VoxelMorph), Cascade (2-cascade VTN), Pyramid (Dual-PRNet++) and our SDHNet. The left and right hemispheres of the brain are combined into one structure for visualization. The horizontal axis denotes all anatomical structures in OASIS: brain stem (BS), vessel (Ve), cerebellum-cortex (CblmC), Inf-Lat-Ventricle (ILV), cerebellum-white-matter (WM), putamen (Pu), caudate (Ca), pallidum (Pa), hippocampus (Hi), Ventral-DC (VDC), 3rd ventricle (3V), 4th ventricle (4V), cerebral-white-matter (CeblWM), lateral-ventricle (LV), thalamus (Th), amygdala (Am), Ac (Accumbens), cerebral cortex (CeblC) and choroid-plexus (CP).

score (1.675 HD95), and even better than 4-cascade VTN with a 80.5% Dice score (1.597 HD95). On the Mindboggle-101 dataset, SDHNet achieves a 63.5% Dice score, which is better than 57.1% of 2-cascade VTN and 60.0% of 3-cascade VTN, and 61.2% of 4-cascade VTN. In terms of HD95, SDHNet still outperforms other methods significantly. For liver CT dataset SLIVER, it is obvious that SDHNet is the best registration method compared with Cascade methods. SDHNet only demands 75% GPU memory (3954MB→2959MB) and 28% inference time (1.14s→0.32s) of 4-cascade VTN, while achieving better registration performance. Therefore, SDHNet is beneficial for the practical situation with limited resources. For SDHNet-*diff*, it still achieves the second-best Dice score and the best Folds compared with Cascade methods with a faster inference speed and a smaller GPU memory. We take images from the LPBA and SLIVER datasets as examples to visualize the results of different methods in Fig. 4 and Fig. 6.

5) Performance on Anatomical Structures: To verify the robust registration performance, we additionally provide boxplots showing the Dice score on each anatomical structure in the OASIS dataset in Fig. 5. We compare our SDHNet with Dual-PRNet++ as the Pyramid method, 2-cascade VTN as the Cascade method, and VoxelMorph as the UNet method. The boxplots show that our SDHNet outperforms other methods on most anatomical structures. In detail, our SDHNet performs better on both large structures such as CeblC or CebIWM and and small structures such as LV, ILV and Hi, which indicates the robust registration performance of our SDHNet.

6) Generalization Performance on LSPIG: We have shown the promising registration performance of our SDHNet on OASIS, Mindboggle-101, LPBA and SLIVER datasets, where the training and testing data have independent and identical distribution. We further evaluate the generalization performance of SDHNet based on the LSPIG dataset. LSPIG contains the pig liver's CT scans, which have the obvious

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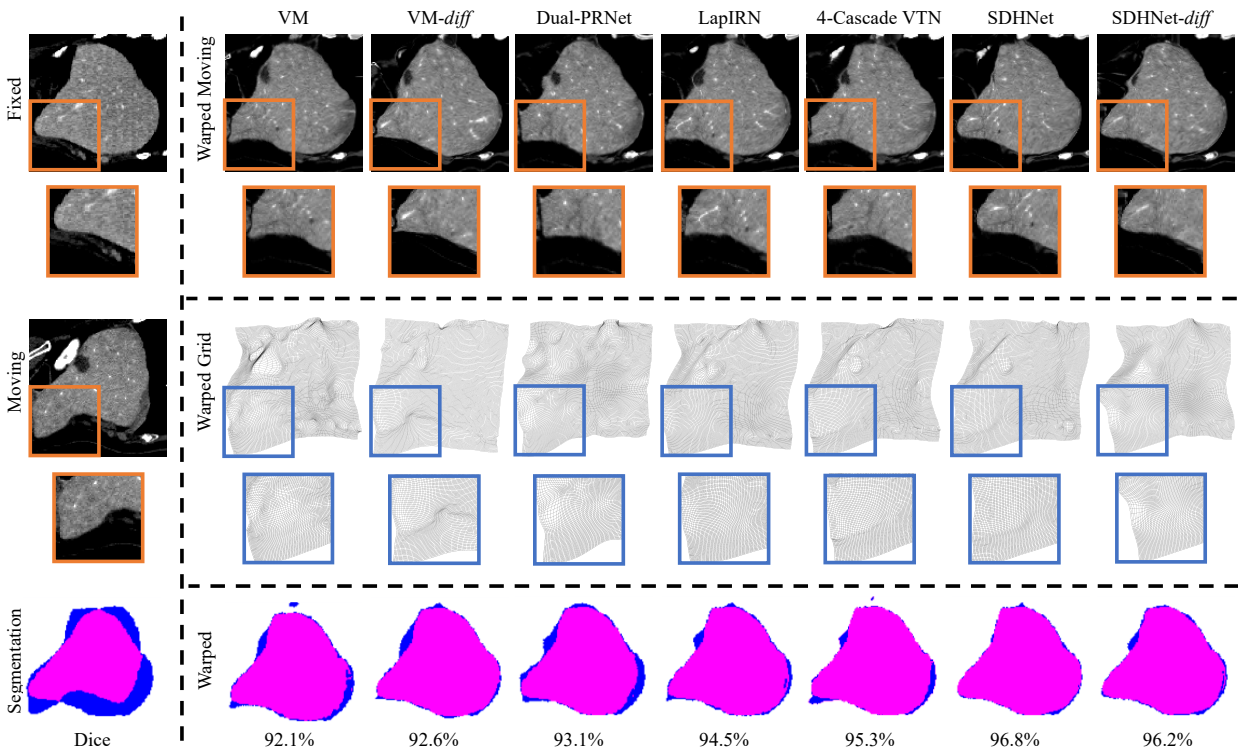


Fig. 6. Visualization of an example in the liver dataset SLIVER. Illustration is consistent with Fig. 4.

TABLE III
COMPARISON RESULTS ON THE SLIVER AND LSPIG DATASET. FOLDS ARE PRESENTED IN E-NOTATION (E.G., 1e-2=0.01). THE STANDARD DEVIATION IS IN PARENTHESES. **BOLD** FONT MEANS THE BEST AND UNDERLINE MEANS THE SECOND-BEST.

Methods	SLIVER		LSPIG	
	Dice (%)	Folds (%)	Dice (%)	Folds (%)
ANTs SyN	90.3 (3.9)	**0**	82.7 (8.4)	**0**
Elastix B-Splines	90.9 (3.8)	1.5e-1	83.0 (6.6)	7.4e-2
Demons	91.4 (3.8)	4.5e-2	83.5 (6.6)	1.1e-2
VoxelMorph	91.2 (2.5)	1.6	83.2 (5.9)	1.1
VoxelMorph-*diff*	91.8 (2.1)	1.7e-2	83.9 (5.5)	3.8e-3
VTN	91.6 (2.3)	1.2	82.5 (6.1)	0.49
Dual-PRNet	92.2 (2.0)	0.63	81.9 (4.6)	0.20
LapIRN-*disp*	93.9 (2.1)	1.8	84.4 (6.4)	1.2
LapIRN	93.5 (2.4)	1.5e-2	84.5 (6.5)	3.7e-3
Dual-PRNet++	94.1 (2.1)	4.4e-1	84.8 (4.6)	1.6e-1
2-cascade VTN	93.5 (1.8)	7.5	84.3 (6.0)	3.1e-1
3-cascade VTN	94.3 (1.7)	4.0e-1	85.0 (6.0)	2.2e-2
4-cascade VTN	94.6 (1.6)	3.1e-1	85.2 (6.4)	1.4e-1
SDHNet	**95.8** (1.6)	3.2e-1	**85.8** (5.7)	1.7e-1
SDHNet-*diff*	95.3 (1.5)	3.0e-3	85.6 (5.8)	3.6e-3

specie difference from human liver datasets such as the liver training dataests MSD and BFH. Take advantage of this specie difference, LSPIG is suitable to evaluate the generalization performance. We apply the model trained on MSD and BFH to LSPIG directly, and the results are shown in Table II. It is obvious that SDHNet along with SDHNet-*diff* achieves the best registration performance in terms of the Dice score and Folds. Specifically, SDHNet improves the Dice score from 83.2% to 85.8% compared with VoxelMorph and from 84.5% to 85.8% compared with LapIRN. SDHNet-*diff* performs a little worse than SDHNet in terms of Dice score but has the second-best Folds number. These results on LSPIG depict the promising generalization performance of our method.

7) Statistical Analysis: We conduct the statistical significance test, and there are statistically significant improvements with p-values$<$0.05 for the Dice score and HD95 in the paired t-test. Specifically, on the OASIS dataset, SDHNet outperforms the second best comparison method 4-cascade VTN with p-values below 5e-5 for the Dice score and 5e-3 for HD95. On the Mindboggle-101 dataset, SDHNet outperforms the second best comparison method Dual-PRNet++ with p-values below 5e-7 in terms for the Dice score and 5e-3 for HD95. On the LPBA and LPBA-v2 datasets, SDHNet outperforms the second-best comparison method DualPRNet++ and 4-cascade VTN with p-values below 5e-5 for the Dice score. On the liver CT datasets such as SLIVER and LSPIG, SDHNet also achieves significant improvements with the p-values below 5e-5 for the Dice score compared with the second best comparison method 4-cascade VTN. The statistical test indicates the significant improvements in registration performance of SDHNet.

B. Ablation Studies

We perform ablation studies for SDHNet on the brain MRI dataset LPBA and liver CT dataset SLIVER at seven aspects: network components, the hierarchical design, the number of network iterations, the weight-sharing scheme, the self-deformation distillation, atals analysis and hyperparameter analysis. We take SDHNet as the example where the impacts are consistent with SDHNet-*diff*.

1) Network Components: We employ six different models (model A to F corresponding first to six rows of Table IV), to clarify the effects of each component in SDHNet. Model A is the plain SDHNet in the first row of Table IV, consisting of a

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TABLE IV
ABLATION STUDIES OF SDHNET ABOUT NETWORK COMPONENTS. **BOLD** FONT MEANS THE BEST.

Fusion		Recurrent		Self-Dist.	Dice (%)	
Adap.	Context	GRU	LSTM		LPBA	SLIVER
					70.4 (1.3)	93.8 (1.7)
✓					71.0 (1.3)	94.3 (1.5)
✓	✓				71.3 (1.2)	94.8 (1.5)
✓	✓	✓			71.7 (1.2)	95.1 (1.5)
✓	✓		✓		71.6 (1.1)	95.1 (1.4)
✓	✓	✓		✓	**72.6** (1.2)	**95.8** (1.6)

TABLE V
ABLATION STUDIES OF SDHNET ABOUT HIERARCHICAL SCALE DESIGN. **BOLD** FONT MEANS THE BEST.

Scale					Dice (%)	
1/2	1/4	1/8	1/16	1/32	LPBA	SLIVER
	✓				70.9 (1.3)	94.3 (1.6)
	✓	✓			72.1 (1.2)	95.4 (1.4)
	✓	✓	✓		72.6 (1.2)	95.8 (1.6)
✓	✓	✓	✓		**73.1** (1.2)	**96.1** (1.6)
	✓	✓	✓	✓	72.6 (1.3)	95.7 (1.4)
✓	✓	✓	✓	✓	**73.1** (1.1)	**96.1** (1.5)

naive parallel generation module and a simple average fusion module. Naive parallel generation module does not have any recurrent structures such as the GRU, while the convolutional layers have comparable parameters with the GRU. Simple average fusion module just adopts the average summation to fuse the hierarchical update deformation fields. Model A does not involve the self-deformation distillation, and is the start point of the following ablation experiments in Table IV.

As mentioned in Section III-C, how to fuse the generated hierarchical deformation fields is a vital problem. Model B adopts the adaptive fusion module rather than the simple average fusion module in the second row of Table IV. The adaptive fusion means the pixel-wise weights in Section III-D are generated only from hierarchical deformation fields themselves without the contextual feature. Comparing model A with model B, the adaptive fusion performs better than the average fusion, indicting its efficiency for the parallel generation. Model C adopts the contextual adaptive fusion in the third row of Table IV, to demonstrate the effect of the contextual feature. The contextual adaptive fusion introduces prior contextual information from the fixed image in addition to deformation fields for generating adaptive weights. From the results, model C outperforms model B and model A, indicting the introduction of contextual information obtains better performance compared with the adaptive fusion and achieve an improvement over the simple average fusion. Furthermore, to explore the interpretability of the contextual adaptive fusion, we visualize the generated weights of three fusion strategies (model A, mode B and model C) in Fig. 7. We take the first deformation direction and 1/4 scale as an example for better visualization. $W_{aver.}$ corresponds to the simple average fusion (model A), $W_{ada.}$ corresponds to the adaptive strategy (model B), and $W_{ada.+cot.}$ corresponds to the contextual adaptive strategy (model C). In Fig. 7, $W_{aver.}$ does not change across different spatial positions, while $W_{ada.}$ and $W_{ada.+cot.}$ vary at different spatial positions. By comparing $W_{ada.}$ and $W_{ada.+cot.}$, it is obvious that details are much more abundant when the contextual information is considered.

We further ablate the effect of recurrent structures in the parallel generation module. Based on model C, model D adopts the GRU and model E adopts the LSTM in the parallel generation module in the fourth and fifth row of Table IV. Results show using recurrent structures including GRU and LSTM can improve the performance, because the long-term connection is considered through the hidden states. Comparing model D with model E, we can find that GRU achieves the comparable performance with LSTM. GRU and LSTM need different learnable weight layers based on the design of gates, where GRU only needs three learnable weights while LSTM needs four learnable weights. Therefore, GRU only demands a 3/4 parameter burden of LSTM and we choose the GRU in our default design. Finally, we ablate the effect of self-deformation distillation. Based on model D, model F adopts the self-deformation distillation, bringing in an improvement without changing any structures in the sixth row of Table IV.

2) Hierarchical Design: SDHNet uses downsampled features to generate three-scale (1/4, 1/8 and 1/6) deformation fields simultaneously in the default setting. We analyse the reasons about downsampled features can generate the full- resolution deformation field with promising performance in three folds: (1) All used hierarchical features are generated from a multi-scale encoder network, encoding plentiful full-resolution information through the numerous feature channels. Contextual features provide guidance from the fixed image for the deformation field generation. (2) Since the feature maps after downsampling consume much less computation to generate a deformation field than before, it is beneficial to introduce more iterations under the same model computation which has been proven helpful for registration [19]. (3) When we use downsampled feature maps to generate the full-resolution deformation field, the upsampling procedure is consistent with the demand of the regularization loss, which can maintain the smoothness of the local region at the full resolution.

Furthermore, we conduct the ablation experiments about the hierarchical design in Table V. From the results, we summarize several insights as following. (1) Merging more scales improves the registration performance. When the scale group is increased from $\{1/4\}$ to $\{1/4, 1/8\}$, the Dice score is improved from 70.9% to 72.1% on the LPBA dataset and from 94.3% to 95.4% on the SLIVER dataset. The performance can be further improved by introducing the 1/16 scale. Different scales correspond to different deformation ranges, and merging more scales helps the network adaptively and simultaneously capture more deformation ranges. (2) Higher scaleachieves the better Dice score, but also brings more model burden (GPU memory and running time). When bringing in the 1/2 scale feature, the Dice score is improved from 72.6% to 73.1% on the LPBA dataest, and the GPU memory is increased from 1764MB to 5404MB and the inference time is increased from 0.24s to 0.98s. Considering the effectiveness of SDHNet, we choose the scale starting from 1/4 in our default setting. It is convenient for SDHNet to increase the scale from 1/4 to 1/2 to address high-precision scenarios. (3) The 1/32 scale does not have an obvious contribution to the registration performance,

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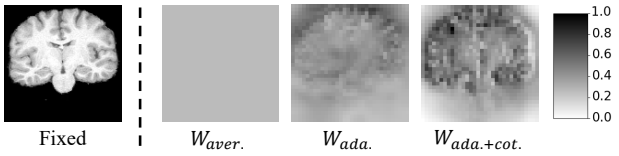


Fig. 7. Weights visualization in different types of fusion strategies.

where the Dice score is similar to that without the 1/32 scale. We reason that the 1/32-scale deformation field or feature corresponds to a too global region, and its influence is covered by the 1/16 scale. Considering 1/32 scale demands more model parameters due to the increment of the generation module, we end the scale at 1/16 in our default setting.

TABLE VI
ABLATION STUDIES OF ITERATIONS IN OUR SDHNET. **BOLD** FONT MEANS THE BEST.

Iterations					Dice (%)	
4	6	8	10	12	LPBA	SLIVER
✓					71.8 (1.1)	94.9 (1.5)
	✓				72.6 (1.2)	95.8 (1.6)
		✓			**72.9 (1.1)**	**96.0** (1.6)
			✓		72.8 (1.1)	**96.0** (1.6)
				✓	**72.9** (1.0)	95.9 (1.5)

3) The Number of Iterations: To explore the impact of the number of iterations in SDHNet, we conduct three experiments with different iterations (termed as Iteration) in Table VII. The results demonstrate that more iterations can generate better registration performance. Specifically, when we increase the number of iterations from 4 to 8, the Dice score is improved from 71.8% to 72.9% on the LPBA dataset and from 94.9% to 96.0% on the SLIVER dataset. Although the improvement can be obtained when the iteration increases from 6 to 8, the relative gain is less than iteration from 4 to 6 while more inference time is demanded. Meanwhile, increasing the number of iterations from 8 to 10 or 12 does not introduce much better performance. We think performance tends to converge when the number of iterations increase, which is also found in the cascaded-based structure such as cascade VTN [19] and RAFT [43]. Considering the effectiveness of SDHNet, we choose the number of iterations as 6 in our default setting, and it is convenient to increase the number of iterations to deal with high precision scenarios.

TABLE VII
ABLATION STUDIES OF SDHNET ABOUT THE WEIGHT-SHARING SCHEME AND THE SELF-DEFORMATION DISTILLATION. **BOLD** FONT MEANS THE BEST.

Experiments			Dice (%)	
			LPBA	SLIVER
	Yes	No		
Sharing	✓		72.3 (1.1)	95.6 (1.5)
		✓	**72.6** (1.2)	**95.8** (1.6)
	$\mathcal{L}_{d-v}$	$\mathcal{L}_{d-g}$		
			71.7 (1.2)	95.1 (1.5)
Self-Dist.	✓		72.3 (1.2)	95.5 (1.7)
		✓	71.8 (1.1)	95.1 (1.6)
	✓	✓	**72.6** (1.2)	**95.8** (1.6)

4) Weight-Sharing Scheme: In our default setting, the modules in different iterations have individual parameters. We further ablate the results when they are weight-sharing in different iterations (termed Sharing) as shown in Table VII. The results show that the weight-sharing modules only drop a small Dice score compared to the individual ones. Meanwhile, it reduces the model parameters from 4.1M to 0.7M, and the performance is still the best compared with the Pyramid or Cascade methods in Table I and Table II which shows that the promising performance is derived from the elaborate design rather than the model parameters. When encountering clinical scenarios with limited storage, the weight-sharing SDHNet is a good choice which has a compact model size but a state-of-the-art performance.

5) Self-Deformation Distillation: Self-deformation distillation scheme applies $\mathcal{L}_{d-v}$ constraints on deformation-value space and $\mathcal{L}_{d-g}$ constraints on deformation-gradient space respectively. We conduct experiments (termed as Self-D.) in Table VII to distinguish their individual effect. The results show that self-deformation distillation with only $\mathcal{L}_{d-g}$ cannot provide the improvement, and the reason is that only spatial-gradient constraints cannot provide an exact optimization direction for the intermediate deformation fields. Only $\mathcal{L}_{d-v}$ can achieve an obvious improvement and $\mathcal{L}_{d-g}$ can be a good complementary to $\mathcal{L}_{d-v}$ with a higher Dice score on both two datasets, which is consistent to the statements in Section III-E.

TABLE VIII
EXPERIMENTS ON DIFFERENT ATLASES IN THE LPBA DATASET. **BOLD** FONT MEANS THE BEST AND UNDERLINE MEANS THE SECOND-BEST.

Methods	Dice (%)		
	Atlas1	Atlas2	Atlas3
ANTs SyN	70.4 (1.5)	71.3 (1.1)	70.7 (1.5)
VoxelMorph	69.0 (1.5)	69.6 (1.0)	68.1 (1.3)
VoxelMorph-*diff*	69.7 (1.5)	70.2 (1.4)	68.6 (1.4)
LapIRN	70.6 (1.5)	71.3 (1.4)	70.3 (1.4)
Dual-PRNet++	71.2 (1.3)	72.1 (1.1)	71.5 (1.5)
4-cascade VTN	70.9 (1.5)	72.3 (1.4)	71.2 (1.3)
SDHNet	**72.6** (1.2)	**73.8** (1.1)	**72.8** (1.3)
SDHNet-*diff*	72.2 (1.3)	73.2 (1.4)	72.4 (1.5)

6) Atlas Analysis: We conduct the atlas-based registration in the LPBA dataset, so the performance may vary depending on the chosen atlas. To verify the robustness against different atlas images, we retrain our SDHNet on two more atlases (the second and third scans) in the LPBA dataset, as shown in Table VIII. These results demonstrate that our performance is consistent and robust against the choice of the atlas, while SDHNet obviously outperforms the other comparison methods.

7) Hyperparameter Analysis: We explain the selection procedure of hyperparameters $\{\lambda, \alpha, \beta\}$. Following VoxelMorph [8], we choose the hyperparameters that perform the best on the validation set. In detail, we empirically set $\lambda = 1$ and $\beta = 1$ and tune parameter α to find the best regularization weight. Then, we fix the best α along the empirical λ and tune parameter β to find the best self-distillation weight. Finally, we tune parameter λ to adjust the self-distillation weight for different intermediate deformation fields. The detailed tuning results on the validation set are shown in Table IX, and we choose $\lambda = 0.8$, $\alpha = 0.5$ and $\beta = 1$ as the default setting.

In addition, we ablate the effects of the regularization weight α and the balance weight λ in self-deformation distillation.

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TABLE IX
HYPERPARAMETERS SELECTION OF SDHNET ON THE VALIDATION SET. **BOLD** FONT MEANS THE BEST

Parameter			Dice (%)	
			LPBA	SLIVER
$\lambda = 1$	$\alpha = 1$	$\beta = 1$	71.9 (1.1)	94.3 (1.5)
$\lambda = 1$	$\alpha = 2$	$\beta = 1$	71.3 (1.2)	93.3 (1.6)
$\lambda = 1$	$\alpha = 0.5$	$\beta = 1$	72.3 (1.2)	94.9 (1.6)
$\lambda = 1$	$\alpha = 0.5$	$\beta = 0.5$	72.6 (1.2)	95.2 (1.6)
$\lambda = 1$	$\alpha = 0.5$	$\beta = 0.1$	72.9 (1.2)	95.5 (1.6)
$\lambda = 0.5$	$\alpha = 0.5$	$\beta = 0.1$	72.5 (1.1)	95.1 (1.5)
$\lambda = 0.8$	$\alpha = 0.5$	$\beta = 0.1$	**73.2** (1.2)	**95.9** (1.4)

TABLE X
ABLATION STUDIES OF THE REGULARIZATION WEIGHT α. FOLDS ARE PRESENTED IN E-NOTATION (E.G., 1e-2=0.01). **BOLD** FONT MEANS THE BEST.

α			LPBA		SLIVER	
0.5	1.0	2.0	Dice (%)	Folds (%)	Dice (%)	Folds (%)
✓			**72.6** (1.2)	1.5e-3	**95.8** (1.6)	3.2e-1
	✓		72.1 (1.1)	8.1e-4	95.0 (1.6)	6.9e-2
		✓	71.4 (1.1)	**1.9e-4**	94.3 (1.4)	**1.5e-2**

As shown in Table X, when we increase the regularization weight α, Folds metric decreases obviously, and our SDHNet still maintains the promising registration performance in terms of the Dice score though it drops as well. Therefore, we can choose the comparative large regularization weight to resist the over-squeezed voxels (Folds). For the balance weight λ, as shown in Table XI, $\lambda = 0.8$ achieves the best performance. As mentioned in Sec. III E, λ should be set in the range of [0, 1] to guarantee that the exponential function is monotonically increasing We think the rationale that $\lambda = 0.8$ obtains the best performance is that, this weight not only treats intermediate deformation fields differently but also does not ignore distillation for the early deformation fields obviously. For λ as 0.5, which distills the early intermediate deformation fields lightly, and λ as 1.0, which treats all intermediate deformation fields equally, the performance is worse than λ as 0.8, which is consistent with our explanation.

VI. CONCLUSION & FUTURE WORK

We propose a novel unsupervised approach SDHNet for deformable image registration. SDHNet uses the recurrent iteration to decompose the large deformation to ease the registration difficulty. In each iteration, SDHNet fully uses hierarchical feature maps by contextual adaptive fusion to handle the complicated regions. Moreover, SDHNet treats the final deformation field as the teacher guidance while adding the self-distillation loss for the intermediate ones, to further boost the registration performance without introducing more inference burden. Experiments on five benchmark datasets demonstrate the superior performance of SDHNet over state-of-the-art methods with a faster inference speed and a smaller GPU memory.

Still, SDHNet has several limitations in its current implementation. First, SDHNet uses the downsampled feature map to estimate the deformation field for maintaining the model efficiency. The introduction of larger feature maps can further improve the registration accuracy, yet at the cost of significantly increased inference burden. Therefore, how to efficiently bring in larger feature maps is one of our future work. Second, SDHNet uses trilinear upsampling to get high-resolution deformation fields, which usually results in blurry boundaries. Therefore, a boundary-aware upsampling module also deserves exploration in the future.

TABLE XI
ABLATION STUDIES OF λ IN THE SELF-DISTILLATION LOSS. **BOLD** FONT MEANS THE BEST

λ			Dice (%)	
			LPBA	SLIVER
0.5	0.8	1.0		
✓			72.1 (1.1)	95.2 (1.5)
	✓		**72.6** (1.2)	**95.8** (1.6)
		✓	72.3 (1.2)	95.4 (1.6)

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